

choptuik_ac_bc · theorems12 ·
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The Depth-Wall Theorem: Window Policy, the Root Anchor, and the $z = 8.56$ Record

z : 1.78 $\rightarrow$ 7.20 $\rightarrow$ 8.56 · [F1]–[F4] topological wall · the 4th echo peak reached

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Series: Monograph No. 11 of the “Twelve
Theorems” cycle

Machine: deep_echo_v21.py +
deep_echo_v22.py · deterministic
PDE machine

Verdict file: einstein_direct/results/
v21_deep_echo.json +
v22_v_budget.json

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Abstract

This monograph records the depth-wall theorem of campaigns v21–v22: the echo-train depth wall of the zoom machine is window-policy sensitive — $z(w = 5) = 1.78 \rightarrow z(w = 3) = 7.20$ with zero junk-gate invasions and exact controls — and the v-budget wall is topological, not budget-sized: the telescope cap never binds, the final stage is vacuum, v-extension is dead code by construction (slack = 0.0 exactly), and the window floor is a coverage ratchet that can only rise.

Forward and late levers are inert or destructive; the root anchor (k1 on zoom 1) plus the coverage-floor anchor break the wall — the depth record $z = 8.56$ (+19%), seven peaks with four on one stage, and the fourth echo peak reached; $\Delta_{\text{eff}} = 0.611 \pm 0.446$ on the quartet is honestly not DSS (3.44 excluded at 6.3σ). The depth budget is recomputed from the reached wall, $S_{\text{req}}(30) \approx 5.0 \times 10^3$. The chain, the mechanism verdicts, the peak ladder, corollaries, honest caveats and the reproduction protocol are included.

Keywords: depth wall, zoom machine, window policy, root anchor, coverage ratchet, echo peaks, $z = 8.56$, machine theorem.

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1. Introduction

Eleventh in the cycle, this monograph records the depth-wall theorem — the mechanism-level result of campaigns v21 and v22 (machines `deep_echo_v21.py` and `deep_echo_v22.py`, LEDGER §§32–33): the echo-train depth wall of the zoom machine is window-policy sensitive, its v-budget variant is topological rather than budget-sized, and a root anchor plus a coverage-floor anchor break it — raising the depth record from $z = 7.20$ to $z = 8.56$ (+19%) and reaching the fourth echo peak.

The theorem is stated as a diagnosis chain: every link is a machine fact with an exact or reproduced control, and the chain as a whole is falsifiable. The user order for v22 — extend the v-budget forward — was tested and honestly redirected by the machine: the lever is backward (anchoring), not forward (`v_ahead`). This monograph fixes both the facts and the redirect, since the redirect is itself a result about the machine's geometry.

2. Notation and Setting

The zoom machine evolves the characteristic solver in stages: each stage marches a window of the double-null grid, then regrid onto a finer window anchored at the deepest covered point. The depth z is measured in log-scale units; Q is the Q -brightness whose peaks mark echoes; v is the retarded-time coordinate of the window edge; the policy parameters are the window factor w , the backward anchor k_{back} , the forward extension v_{ahead} , and the anchor-from-zoom switch. The v21 protocol wraps the march with a q7 junk gate on active rows, stage rollback with window tightening on invasion, and per-echo checkpoints.

The setting of the theorem is the v21 control chain ($\text{eps} = 10^{-3}$ above A^* , $n = 800$) and its v22 diagnosis: reproduce the chain with zero deviation, probe the wall mechanism by mechanism, then probe window-placement policies. All probes are deterministic and reported in `results/v21_deep_echo.json` and `results/v22_v_budget.json`.

3. Statement of the Theorem

Theorem 11 (the depth-wall chain). For the controlled zoom machine of the programme the following chain of facts holds:

Table 11.1. The depth chain ($v_{21} \rightarrow v_{22}$)

Link	Operation	Depth z	Peaks	Stop
$z(w = 5)$	protocol defaults	1.78	1	singularit y
$z(w = 3)$	window $w = 3$	7.20	3	v_{exhaust} ed
$z(w = 1.5)$	window $w = 1.5$	6.66	3	v_{exhaust} ed
$z(\text{root } k_1 = 0.60)$	root anchor + coverage-floor anchor	8.56	2	v_{exhaust} ed
$z(\text{root } k_1 = 0.52)$	root anchor + coverage-floor anchor	6.99	7 (4 on one stage)	v_{exhaust} ed

(i) the wall is window-policy sensitive: $z(w = 5) = 1.78 \rightarrow z(w = 3) = 7.20$ with zero junk-gate invasions, and the controls are exact (plain ZoomRunner $n = 800$ reproduces $z = 1.782$; $n = 600$ reproduces the v_{20a} verdict 4.598) — the wrapper is neutral, the lever is the window;

(ii) the v -budget wall is topological, not budget-sized: [F1] the telescope cap never binds (4/4 zooms free, max ahead/slack = 0.295); [F2] the final stage is vacuum ($Q \sim 0$, $w_{\text{cells}} = 0$, $mx = 0$ on 100% of 799 rows — the trigger ladder descends in v , $0.4781 \rightarrow 0.4072 \rightarrow 0.4035 \rightarrow 0.4015$, while the window looks ahead); [F3] v -extension is dead code by construction (slack = 0.0 exactly, 1/1 traced call); [F4] the window floor is a coverage ratchet — it can only rise (floor $\equiv 0.40113$ across the whole ladder);

(iii) forward and late levers are inert or destructive: $v_{\text{ahead}} = 3.0$ is inert (clamped by the buffer); the global backward anchor is destructive (the restart row lands in the incoming-pulse region, $v_p = 0.5 \pm 0.1$, $z \sim 1.5\text{--}1.9$); every late anchor is clamped by the ratcheted floor (late = 1/2/3 from zoom 3: $z = 7.20$ in all three, the exact control);

(iv) the root anchor + coverage-floor anchor break the wall: k_1 on zoom 1 lowers the root floor below the expected echo, and from zoom 2 the window anchors at the deepest covered point — the floor freezes and the ladder descends into covered territory: $k_1 = 0.60 \rightarrow$

$z = 8.56$ (record, +19%); $k_1 = 0.52 \rightarrow 7$ peaks with 4 on one stage and the fourth peak reached; $k_1 = 0.80 \rightarrow 3.96$ (singularity — the anchor has a working range); the new wall is the root floor;

(v) the depth budget is recomputed from the reached wall: $S_{\text{req}}(30) = e^{(\gamma \cdot 20.65)} \approx 5.0 \times 10^3$ ($\times 10$ margin 5.0×10^4) from the $z = 7.20$ wall — the v20b budget updated by the machine's own progress.

4. Sketch of the Machine Proof

4.1. Reproduction and diagnosis

The v22 diagnosis opens by reproducing the v21 $w = 3$ chain with deviation $0.0e+00$ (the fork is neutral), then traces the four mechanisms [F1]–[F4] directly on the chain data: the telescope accounting per zoom, the vacuum fraction of the final stage, the traced slack of `_try_extend` (zero by construction, since the call happens at $j \geq n$ where $v_{\text{now}} = v[n-1] = \text{stage_v}[1]$), and the floor history across the ladder (a monotonically rising ratchet). Each mechanism is a falsifiable statement about the stage logs, and each is reported with its counter.

4.2. Policy probes and the redirect

The probe battery tests the lever directions: forward (`v_ahead`), global backward, late backward, and the root/coverage pair. The redirect is data-driven: the destructive global anchor shows where the restart row must not land; the clamped late anchors show that only the root floor can be lowered; the working range $k_1 \in \{0.52\text{--}0.60\}$ shows the anchor's window of validity, with $k_1 = 0.80$ overshooting into a singularity. The deep run on the accepted policy ($k_1 = 0.52$, anchor from zoom 2) reaches $z = 6.99$ with 7 peaks and delivers the fourth peak on stage 4.

Table 11.2. Mechanism verdicts [F1]–[F4] and probes (v22)

Fact	Verdict	Counter
[F1] telescope cap	never binds	4/4 zooms free; max ahead/slack = 0.295

Fact	Verdict	Counter
[F2] final stage	vacuum	$Q \sim 0$, $w_cells = 0$, $mx = 0$ on 100% of 799 rows
[F3] v-extension	dead code by construction	slack = 0.0 exactly (1/1 traced)
[F4] window floor	coverage ratchet (only rises)	floor $\equiv 0.40113$ across the ladder
[C2] global backward anchor	destructive	restart in incoming pulse; $z \sim 1.5\text{--}1.9$
[C2b] late anchor	clamped by the floor	late = 1/2/3: $z = 7.20$ (control)
[C2c] root + floor anchor	breaks the wall	$k1 = 0.60 \rightarrow 8.56$; $k1 = 0.52 \rightarrow 7$ peaks

5. Numerical Results

Table 11.3. The deep run ($k1 = 0.52$, anchor from zoom 2): peak ladder

Stage (zoom)	Peaks	v-peaks	Q-peaks (order)
zoom 0	1	0.41512	7.05×10^{-1}
zoom 1	0	—	—
zoom 2	1	0.40042	5.6×10^{-6}
zoom 3	1	0.39863	1.3×10^{-9}
zoom 4	4	0.39920 / 0.39941 / 0.39945 / 0.39952	$10^{-12}\text{--}10^{-11}$ (quartet)
zoom 5	0	—	—

The quartet of zoom 4 delivers $\Delta_{\text{eff}} = 0.6114 \pm 0.4459$ — honestly not DSS (the literature period 3.44 is excluded at 6.3σ ; the estimate is compatible with zero); the DSS verdict and its consequences belong to monograph No. 12. For the wall theorem the relevant fact is structural: the fourth peak is reached at all — the root-anchor policy carries the ladder into territory that was walled off at 7.20.

$$S_{\{req\}}(30) \approx 10^{20.65} \approx 10^3 \text{ (margin)} 10^4 \quad (11.1)$$

6. Corollaries

Corollary 11.1 (the wall taxonomy). The programme’s depth walls are now classified: the junk wall (q7 gate — zero invasions on the accepted policies), the window wall (policy-sensitive, broken by $w = 3$), the v-budget wall (topological [F1]–[F4], broken by root anchoring), and the root floor (the current wall). Each wall has its own falsifiable signature, and the v23 frontier is defined against the last of them.

Corollary 11.2 (the lever is backward). The user hypothesis of v22 (forward v-budget) is honestly redirected: forward extension is dead code by construction and the global backward anchor is destructive; only the root anchor lowers the correct floor. This redirect is a machine fact about the zoom geometry, reusable in any future regridding design.

Corollary 11.3 (budget discipline). The budget (11.1) is recomputed from the reached wall, not from optimism: every policy upgrade in the chain $1.78 \rightarrow 7.20 \rightarrow 8.56$ moved $S_{\text{req}}(30)$ by a computable factor, and the v23 protocol inherits a quantified target rather than an open one.

7. Honest Caveats

The junk-gate wrapper [P2]/[P3] is not connected in the v22 probes: the anchoring results are pre-protocol facts, and a full junk-gated run on the adopted policy is exactly the v23 order. The working range of the root anchor is bracketed but not swept finely ($k_1 \in \{0.52, 0.60, 0.80\}$); a fine k_1 sweep (0.50–0.70) is part of v23. The fourth-peak quartet is contaminated by the floor region (Q at 10^{-12} – 10^{-11} levels); its period is honestly not DSS and may be a sub-echo structure. The depth record 8.56 is machine- and policy-specific: it certifies that the wall was topological, not that the physics knows a barrier there. All chains are deterministic and reproducible; the deviation of the v21 reproduction is 0.0e+00.

8. Reproducibility

Table 11.4. Reproduction protocol for Theorem 11

Step	Command	Output
1	<code>python3 deep_echo_v21.py --phase all</code>	<code>results/v21_deep_echo.json</code> (~3 min; chain + probe + ladder)

Step	Command	Output
2	<code>python3 deep_echo_v22.py --phase all</code>	results/v22_v_budget.json (~5–7 min; cert + probes + deep)
3	<code>python3 -m pytest tests/ -q</code>	58 passed (v21/v22 tests included)

The machines are deterministic (BLAS = 1, fixed seeds) and run in minutes on a laptop; in Termux expect roughly 2–3× the laptop time. The data correspond to commit 5b256cd; self-contained copies live in `einstein_direct/open9/machines/`.

References

- [1] Repository `wild8highlander/choptuik_ac_bc`: `einstein_direct/deep_echo_v21.py`;
`einstein_direct/results/v21_deep_echo.json` (campaign v21, LEDGER §32).
- [2] Ibid.: `einstein_direct/deep_echo_v22.py`; `einstein_direct/results/v22_v_budget.json` (campaign v22,
LEDGER §33).
- [3] Ibid.: `einstein_direct/open9/README.md` (the Section 5 protocol reference).
- [4] Ibid.: `einstein_direct/README.md` §§3.10–3.11.
- [5] Monographs No. 3, No. 12 of the theorems12 cycle.